The motivation paragraph: three sentences, three decisions

For Halil, 2026-08-29

What the search found

A better source exists and it is recent. It also turned up a second problem in the same paragraph that we had not looked at.

The paragraph is `docs/paper/sections/intro.tex` lines 14 to 22. It makes three claims and each needs a separate decision. Every proposed sentence below is a rewrite, so `check_subseq.py` flags all of them and none is mine to apply.

Sentence 1, line 15. The separation itself

Current: "its privacy risk is concentrated at training time~[**?**, **?**]".

The claim is fine. The citations can be improved and reduced, which is what the PI asked for on 2026-08-21.

Bai, Hu, Ye, Li, Wang and Xu, *Membership Inference Attacks and Defenses in Federated Learning: A Survey*, ACM Computing Surveys 57(4), Article 89, 2025, DOI 10.1145/3704633, states the separation as a table row. Centralised learning is attacked at the inference phase and the adversary knows the target model. Federated learning is attacked at the training phase and the adversary knows the target model and its historical versions.

Proposal. Replace the two citations with this one. Two keys become one, and the survey is the authority the sentence needs. Nothing else changes.

Sentence 2, line 17 to 18. Gradient reconstruction

Current: "A gradient permits reconstruction of the batch that produced it~[**?**]."

This is a correctness problem and it is new. Du, Hu, Wang, Sun, Gong, Ren and Chen, *SoK: On Gradient Leakage in Federated Learning*, USENIX Security 2025, contradicts the sentence as written. From their abstract, gradient inversion is "notably constrained, fragile, and easily defensible", and "even simple post-processing techniques applied to gradients can serve as effective defenses". Their Theorem 4.3 proves a dimension bound. When the parameter count is smaller than the batch size times the input dimension, distinct inputs produce identical gradients, so reconstruction is not merely hard but impossible.

Our sentence states the reconstruction unconditionally and against a semi-honest adversary. A 2025 SoK at a top venue says that is wrong in that setting.

Two ways out.

Name the actor, which is the honest fix and also the stronger claim. A server that modifies the model it distributes reconstructs the batch, and does so even through secure aggregation across a hundred clients. That is LOKI, Zhao et al., IEEE S&P 2024, which reports leaking 76 to 86 per cent of samples in a single round where prior work leaks under one per cent. The cost is that this adversary deviates from the protocol, and our Section IV assumes a semi-honest server, so the sentence would motivate the design with an adversary we do not defend against. If we take this route the sentence must say the server deviates.

Or delete the sentence and let the membership number carry the paragraph.

Recommendation. Delete it. It is one sentence, the paragraph does not need it, deletion is permitted by the subsequence rule, and it removes a claim a reviewer can refute with a 2025 SoK. Sentence 3 does the work.

One clause worth having somewhere, later. The same SoK finds that late-training models resist inversion while early ones do not. The single artifact we expose is a fully locally trained displacement, which is the late kind. That is an argument in our favour and it belongs in the report, hedged, because it is an inference from their setting to ours.

Sentence 3, lines 19 to 21. The numbers

Current: "An observer of the per-round updates mounts membership inference at 87% accuracy on one model and dataset, where the same attack against the final model alone falls to 54.5%."

Both numbers are wrong for what the sentence says, as established. Note that no pure number substitution repairs it, because the phrase "the same attack" is the false part, so every option here is a rewrite.

Option A, stay with Nasr and use its tables

"An observer of the per-round updates mounts membership inference at 79% accuracy on one model and dataset, where a black-box attack against a fully trained model reaches 68%."

Both figures come from tables of the cited paper rather than its introduction. 79.2 per cent is the passive global attacker, Table X, which is what the word observer means. 67.7 per cent is Table VIII, the stand-alone black-box cell, which is the number their own tables give for the comparison their introduction puts at 54.5.

Honest, and the smallest change. The contrast falls from thirty-three points to eleven, which is a weak motivation for a paper whose whole design follows from it.

Option B, use the recent work, which is what the search was for

Zhu, Li, Gu, Yao, Fan and Han, *FedMIA*, CVPR 2025, pages 20643 to 20653, DOI 10.1109/CVPR52734.2025.01922. Its Table 1 carries both halves of our contrast in one run, one metric, one target.

"An adversary that aggregates a client's per-round updates mounts membership inference at a true positive rate of 25.3% at a false positive rate of 0.1%, where the same statistic read from a single model snapshot reaches 0.18%."

AlexNet on CIFAR-100, ten clients, three hundred rounds. The ResNet-18 row is 16.82% against 0.36% if you prefer the more familiar architecture.

Three reasons this is better. Both numbers come from the same table, the same run and the same statistic, so the comparison is genuinely like for like, which is exactly what Nasr cannot give us. The metric is true positive rate at a low false positive rate, which is what the field has used since Carlini et al. 2022 and which a reviewer now expects. And the contrast is two orders of magnitude rather than eleven points.

One qualification, and it changes the wording rather than the argument. The single-round figure is a model snapshot taken during training, not a separately released final model. So the defensible sentence is that the round sequence leaks far more than any one snapshot of the same run. It is not a statement about a released final model. The proposed wording above already says snapshot.

A second qualification in our favour. That 0.18% is the best of thirty observed snapshots, not an average one, so the gap is a conservative lower bound.

Recommendation. Option B. You asked for a more recent work and this is a better one on every axis that matters.

What no citation supports, and what we should therefore not write

The conclusion that exchanging a single contribution once removes most of the attack surface is ours. Nothing measures the residual risk of a one-shot exchange against a multi-round one on the same task. The surveys assert the principle and hedge it, and the IJCAI 2025 one-shot survey writes “potentially achieves even stronger security”.

So state the antecedent, which is well supported, and derive the design choice from it. Do not attribute the conclusion to a citation. Our protocol also encrypts the single contribution, so the one-shot property is not what protects confidentiality against the server. It bounds the size of the surface. Keeping those two arguments apart is what will survive a reviewer from the TDSC pool.

Summary of the three decisions

Sentence 1: swap two citations for the ACM survey. Reduces the count by one.

Sentence 2: delete it, or name a deviating server and cite LOKI.

Sentence 3: Option A or Option B. I recommend B.